Logarithms Assessment

MC1) $\log a + \log b - \log c =$

A) $\log(a+b+c)$ B) $\log(ab-c)$

C) $\log\frac{ab}{c}$ D) $-\log(abc)$

MC2) If $2^{x+1} = 10$, which one of the following is **incorrect**?

A) $2^x = 5$ B) $x\log 2 = \log 10$

C) $(x+1)\log_3 2 = \log_3 10$ D) $\log 2 + \log 2^x = \log 10$

MC3) $\log(a+b) + \log(c) - \log(d) =$

A) $\log(\frac{abc}{d})$ B) $\log(\frac{a+b+c}{d})$

C) $\log(\frac{(a+b)c}{d})$ D) $\log((a+b)+\frac{c}{d})$

MC4) If $a^{x+1} - a^x = 1$ then $x =$

A) $-\log_a(a-1)$ B) $-\frac{1}{a}$

C) 0 or -1 D) 1

Q1.

A student was asked to give the exact solution to the equation

$$2^{2x+4} - 9(2^x) = 0$$

The student's attempt is shown below:

$$2^{2x+4} - 9(2^x) = 0$$

$$2^{2x} + 2^4 - 9(2^x) = 0$$

$$\text{Let} \quad 2^x = y$$

$$y^2 - 9y + 8 = 0$$

$$(y-8)(y-1) = 0$$

$$y = 8 \text{ or } y = 1$$

$$\text{So } x = 3 \text{ or } x = 0$$

(a) Identify the two errors made by the student.

(2)

(b) Find the exact solution to the equation.

(2)

Q2.
Use logarithms to solve the equation:

$$3^{2x+1} = 4^{100}$$

Give your answer correct to 3 significant figures. Fully justify your answer. **[4]**

Q3.
(i) Write down the value of $\log_6 36$.

(1)

(ii) Express $2\log_a 3 + \log_a 11$ as a single logarithm to base a.

(3)

Q4.
Given that $0 < x < 4$ and

$$\log_5(4 - x) - 2\log_5 x = 1,$$

find the value of x.

(6)

Q5.
(a) Given that

$$2\log_3(x-5) - \log_3(2x-13) = 1,$$

show that $x^2 - 16x + 64 = 0$.

(5)

(b) Hence, or otherwise, solve $2\log_3(x-5) - \log_3(2x-13) = 1$.

(2)

Q6.
Use logarithms to solve the equation: $3^{2x+1} = 5^{200}$ .Give your answer to 3 significant figures.

(5)

Q7.
Solve the equation $2^{4x-1} = 3^{5-2x}$, giving your answer in the form $x = \frac{\log_{10} a}{\log_{10} b}$

(6)

Total: 40 marks